King Abdulaziz University

College of Engineering

Electrical and Computer Engineering

EE 424 (Antenna)

Assignment L14

(Aperture [Horn] Antenna)

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Requirements: design a pyramidal horn antenna at 1.2GHz and studying the effects of varying the aperture size three versions as required and comparing the results to understand the effect of size.

Starting with calculations to calculate aperture dimensions:

f= 1.2GHz lamda= 0.25m

a=0.162m

b=0.081m

Small aperture:

W=0.5m H=0.25m Flare length=0.125m

Medium aperture:

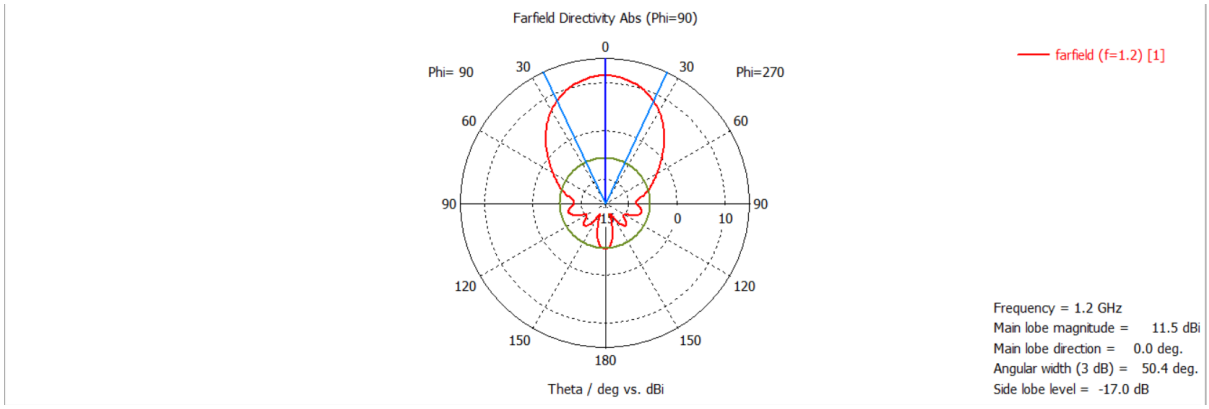
W=0.75m H=0.375m Flare length=0.281m

Large aperture:

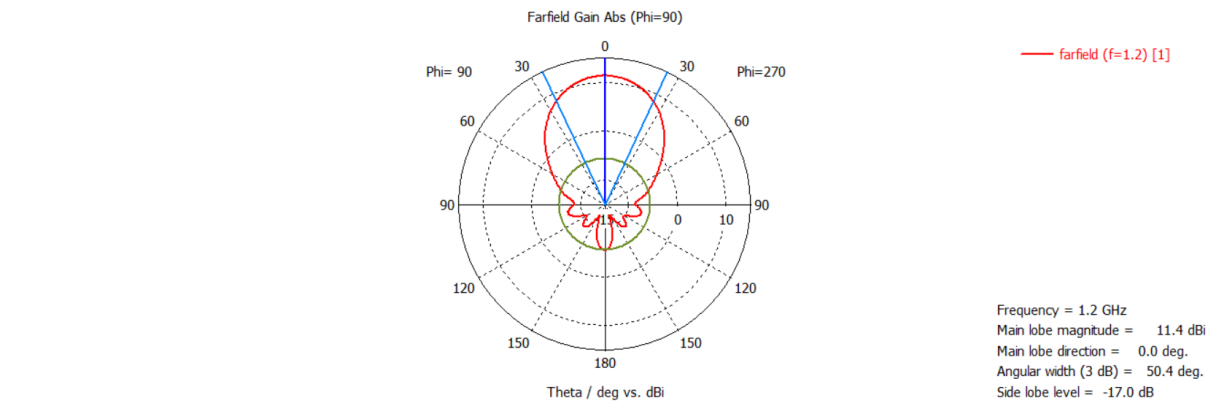
W=1m H=0.5m Flare length=0.5m

All the calculations performed by hand on last page**

Starting with small aperture the gain and directivity was 11.5dBi with beamwidth 50.4 deg.

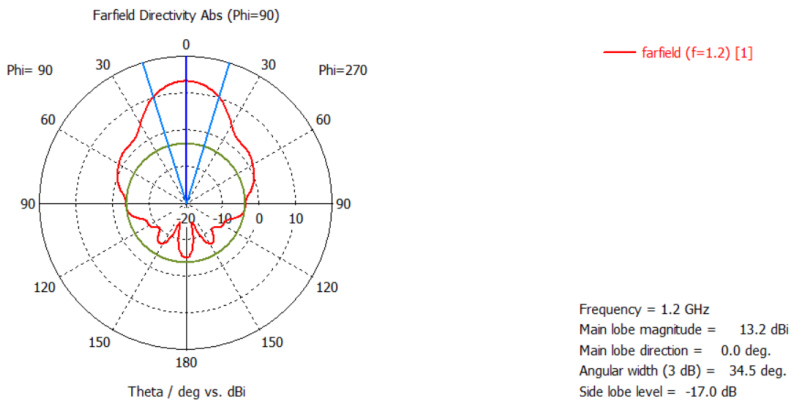


Small aperture directivity

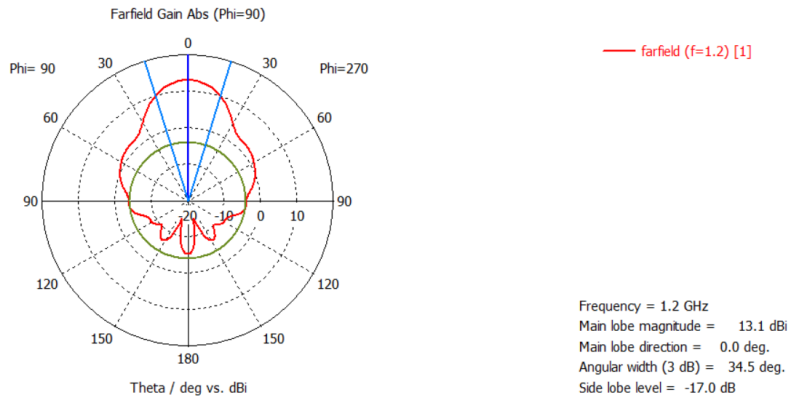


Small aperture Gain

Enlargement of the Aperture antenna to medium size improved gain and directivity to be 13.2 dBi and lowers the beamwidth to 34.5 deg as shown:

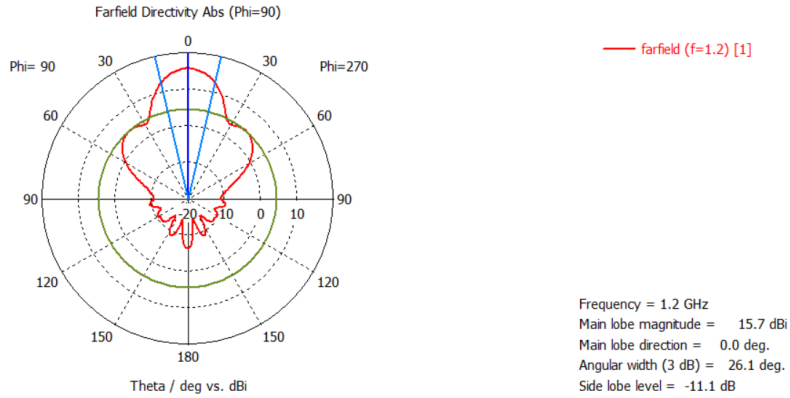


Medium aperture directivity

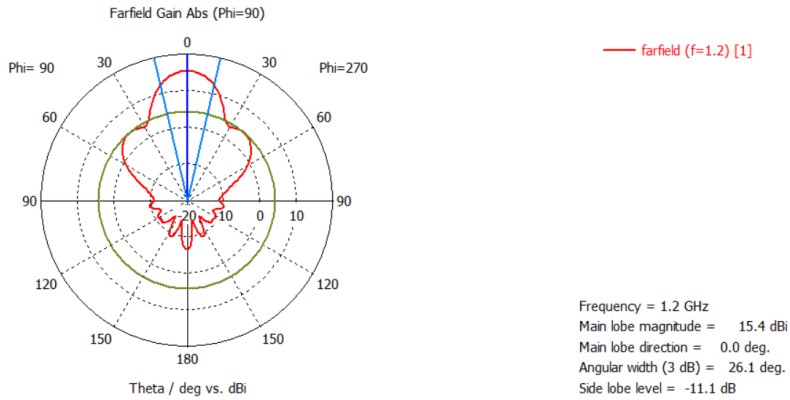


Medium aperture Gain

Making Aperture antenna to be large size improved gain and directivity to be 15.4 dBi, 15.7dBi and lowers the beamwidth to 26.1 deg as shown:



Large aperture directivity



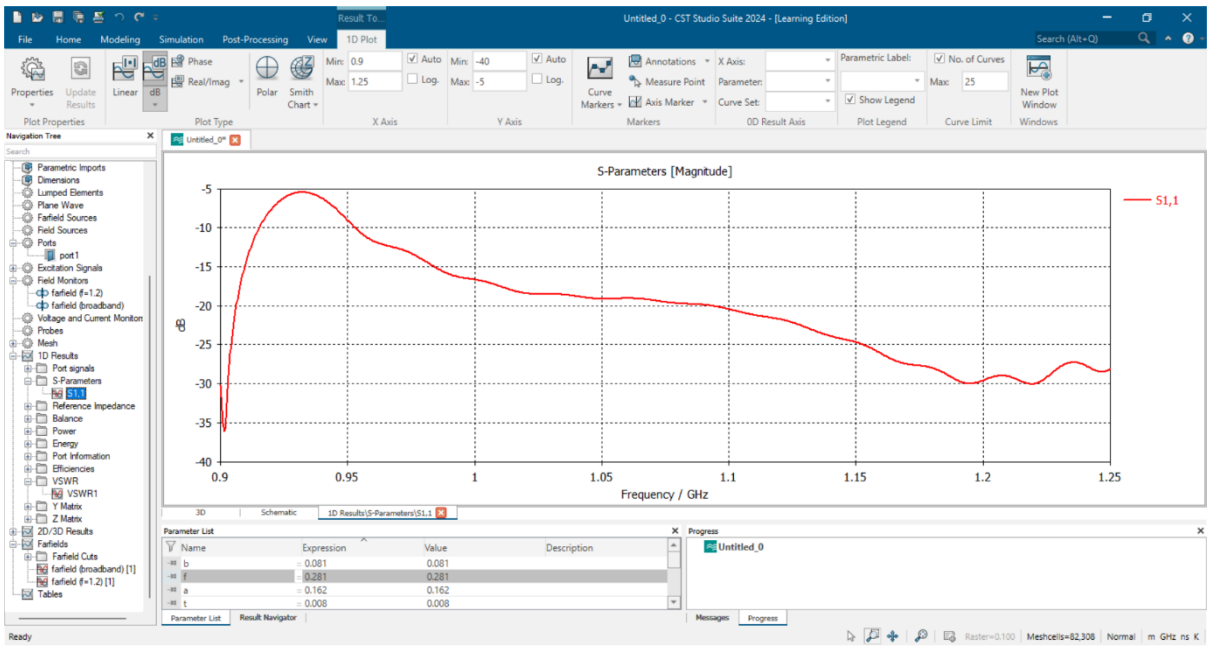
Large aperture directivity

Analysis and Comparing

As the simulated results shown, the gain and directivity improved by aperture enlargement and compering with dipole antenna horn antenna gives more gain and better beamwidth to be more directive on specific direction.

As we learn in class aperture antenna produces a directional beam so its expected to gives high gain.

Doing more studies to see if S11 will be affected, it did but the effect wasn't that bad it gives between -20dBi to -30dBi.



S11 Medium aperture

References:

1-Lecture 14

2-Given formulas from the file from given instructor

3-Hand calculations:

14

1.2GHz

$f_c = \frac{1.2}{1.3} = 0.923\ GHz$

$f_c = \frac{3\times10^8}{2a}$

$\lambda = \frac{c}{f} = \frac{3\times10^8}{1.2} = 0.25\ m$

$a = \frac{3\times10^8}{2\times0.923\times10^9} = 0.162\ m$

$b = \frac{a}{2} = \frac{0.162\ m}{2} = 0.081\ m$

Small

$W = 2\lambda = 0.5\ m$ $H = 1\lambda = 0.25\ m$ $A = W\times H = 0.125\ m$

medium

$W = 3\lambda = 0.75\ m$ $H = 0.375\ m$ $A = 0.281\ m$

Large

$W = 4\lambda = 1\ m$ $H = 0.5\ m$ $A = 0.5\ m$

Flare length $\gamma = 0.25\lambda = 0.0625\ m$

Small $\rho = \frac{h^2}{8\gamma} = \frac{0.25^2}{8\times0.0625} = 0.125\ m$

medium $\rho = \frac{h^2}{8\gamma} = \frac{0.375^2}{8\times0.0625} = 0.281\ m$

Large $\rho = \frac{h^2}{8\gamma} = \frac{0.5^2}{8\times0.0625} = 0.5\ m$